

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

I M. VAISHNAV (192524251) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. VAISHNAV (192524251) .

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%	5	2	1	1
Identification of Key Issues	20%	1	1	1	1
Application of Concepts	25%	1	1	1	1
Decision Making	20%	1	1	1	1
Clarity of Response	15%	1	1	1	1

24/50 

		g of the scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned , logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganize d and difficult to understand

Smart campus (192.168.100.0/24):-

Q1.

Department	Subnet	Network address	Host Range	Broadcast add.	Usable hosts
School of computing	125	192.168.100.0	192.168.100.1 192.168.100.126	192.168.100.127	126
School of electronics	126	192.168.100.128	192.168.100.129 192.168.100.190	192.168.100.191	62
School of Mechanical Engineering	127	192.168.100.192	192.168.100.193 192.168.100.222	192.168.100.223	30

Remaining unused IP Range:-

192.168.100.224 - 192.168.100.255.

Q2.

Software company (10.10.0.0/16):-

Department	Subnet	Network add.	Host Range	Broadcast add.	Usable hosts
Software Development	124	10.10.0.0	10.10.0.1- 10.10.0.254	10.10.0.255	254
Quality Assurance	125	10.10.1.0	10.10.1.1- 10.10.1.126	10.10.1.127	126
Human Resource	126	10.10.1.128	10.10.1.129- 10.10.1.190	10.10.1.191	64
Executive Management	127	10.10.1.192	10.10.1.193- 10.10.1.222	10.10.1.223	30

unused IP Range :-

10.10.1.224 - 10.10.255.255.

Q3. IPv6:

given address:-

2001:0db8:abcd:0000:0000:0000:0000:0000

1. compressed form

2001:db8:abcd::164

2. Expanded Form

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network prefix (164)

2001:0db8:abcd:0000::164

4. hosts supported

2^{64}

5. Total possible host addresses

18,446,744,073,709,551,616.

Q4.

Routing Table:-

Routing Table

• 192.168.1.0/24

• 192.168.2.0/24

• 192.168.3.0/24

Destination IP : 192.168.2.45

1. Destination Subnet

2. 192.168.2.0/24

3. Matching Routing Entry

4. 192.168.2.0/24

5. Next Hop Network / Interface.

5. Default Route (if no match)

. 0/0

Q5.

Layer-3 Router:-

Administration LAN (192.168.10.0/26)

- Network Add.: 192.168.10.0
- Host Range: 192.168.10.1 - 192.168.10.62
- Broadcast add.: 192.168.10.63
- Gateway: 192.168.10.1

Three Computer IPs:-

- 192.168.10.2
- 192.168.10.3
- 192.168.10.4

Finance LAN (192.168.10.64/26)

- Network address: 192.168.10.64
- Host Range: 192.168.10.65 - 192.168.10.126
- Broadcast add.: 192.168.10.127
- Gateway: 192.168.10.65

Three computer IPs:-

- 192.168.10.66
- 192.168.10.67
- 192.168.10.68